

Homework 4 (1.10.4 ~ 2.7.1)

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1.10.4

We start from Eq. (1.10.55)

$$|\psi(t)\rangle = \sum_{m=1}^{\infty} |m\rangle \langle m|\psi(0)\rangle \cos \omega_m t, \quad \omega_m = \frac{m\pi}{L}$$

$$\psi(x,t) = \langle x|\psi(t)\rangle = \sum_{m=1}^{\infty} \langle x|m\rangle \langle m|\psi(0)\rangle \cos \omega_m t$$

$$\langle x|m\rangle = \psi_m(x) = \left(\frac{2}{L}\right)^{\frac{1}{2}} \sin \frac{m\pi x}{L} \quad (\text{Eq. (1.10.50)})$$

$$\begin{aligned} \langle m|\psi(0)\rangle &= \int_0^L \left(\frac{2}{L}\right)^{\frac{1}{2}} \sin \frac{m\pi x}{L} \cdot \psi(x,0) dx \\ &= \left(\frac{2}{L}\right)^{\frac{1}{2}} \left[\int_0^{\frac{L}{2}} \frac{2xh}{L} \sin \frac{m\pi x}{L} dx + \int_{\frac{L}{2}}^L \frac{2h}{L} (L-x) \sin \frac{m\pi x}{L} dx \right] \\ &\equiv \left(\frac{2}{L}\right)^{\frac{1}{2}} [I + II] \end{aligned}$$

$$\begin{aligned} I &= \int_0^{\frac{L}{2}} \frac{2xh}{L} \sin \frac{m\pi x}{L} dx \\ &= -\frac{2h}{L} \cdot \frac{L}{m\pi} \int_0^{\frac{L}{2}} x d \cos \frac{m\pi x}{L} \\ &= -\frac{2h}{L} \cdot \frac{L}{m\pi} x \cos \frac{m\pi x}{L} \Big|_0^{\frac{L}{2}} + \frac{2h}{L} \cdot \frac{L}{m\pi} \int_0^{\frac{L}{2}} \cos \frac{m\pi x}{L} dx \\ &= -\frac{2h}{m\pi} \cdot \frac{L}{2} \cos \frac{m\pi}{2} + \frac{2h}{L} \left(\frac{L}{m\pi}\right)^2 \sin \frac{m\pi x}{L} \Big|_0^{\frac{L}{2}} \\ &= -\frac{hL}{m\pi} \cos \frac{m\pi}{2} + \frac{2h}{L} \left(\frac{L}{m\pi}\right)^2 \sin \frac{m\pi}{2} \end{aligned}$$

$$\begin{aligned} II &= \int_{\frac{L}{2}}^L \frac{2h}{L} \cdot L \sin \frac{m\pi x}{L} dx - \int_{\frac{L}{2}}^L \frac{2hx}{L} \sin \frac{m\pi x}{L} dx \\ &= -2h \cdot \frac{L}{m\pi} \cos \frac{m\pi x}{L} \Big|_{\frac{L}{2}}^L + \frac{2h}{L} \cdot \frac{L}{m\pi} \int_{\frac{L}{2}}^L x d \cos \frac{m\pi x}{L} \\ &= \frac{2hL}{m\pi} \cos \frac{m\pi}{2} - \frac{2hL}{m\pi} \cos m\pi + \frac{2h}{L} \cdot \frac{L}{m\pi} x \cos \frac{m\pi x}{L} \Big|_{\frac{L}{2}}^L - \frac{2h}{m\pi} \int_{\frac{L}{2}}^L \cos \frac{m\pi x}{L} dx \\ &= \frac{2hL}{m\pi} \cos \frac{m\pi}{2} - \frac{2hL}{m\pi} \cos m\pi + \frac{2hL}{m\pi} \cos m\pi - \frac{hL}{m\pi} \cos \frac{m\pi}{2} - \frac{2h}{m\pi} \cdot \frac{L}{m\pi} \sin \frac{m\pi x}{L} \Big|_{\frac{L}{2}}^L \\ &= \frac{hL}{m\pi} \cos \frac{m\pi}{2} - \frac{2h}{L} \left(\frac{L}{m\pi}\right)^2 \sin m\pi + \frac{2h}{L} \left(\frac{L}{m\pi}\right)^2 \sin \frac{m\pi}{2} \\ &\quad 0 \end{aligned}$$

$$I + II = \frac{4h}{L} \left(\frac{L}{m\pi}\right)^2 \sin \frac{m\pi}{2}$$

$$\begin{aligned} \text{Therefore, } \psi(x,t) &= \sum_{m=1}^{\infty} \left(\frac{2}{L}\right)^{\frac{1}{2}} \sin \frac{m\pi x}{L} \cdot \left(\frac{2}{L}\right)^{\frac{1}{2}} \cdot \frac{4h}{L} \left(\frac{L}{m\pi}\right)^2 \sin \frac{m\pi}{2} \cos \omega_m t \\ &= \sum_{m=1}^{\infty} \sin \left(\frac{m\pi x}{L}\right) \cos \omega_m t \cdot \left(\frac{8h}{\pi^2 m^2}\right) \sin \left(\frac{\pi m}{2}\right) \end{aligned}$$

2.1.1

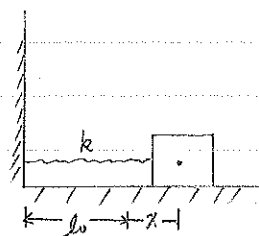
$$T = \frac{1}{2} m \dot{x}^2$$

$$V = \frac{1}{2} k x^2$$

$$L = T - V = \frac{1}{2} m \dot{x}^2 - \frac{1}{2} k x^2$$

$$\frac{\partial L}{\partial \dot{x}} = m \dot{x}, \quad \frac{\partial L}{\partial x} = -kx$$

$$\frac{d}{dt} \left(\frac{\partial L}{\partial \dot{x}} \right) - \frac{\partial L}{\partial x} = 0 \Rightarrow m \ddot{x} + kx = 0.$$



2.1.2

$$T = \frac{1}{2} m \dot{x}_1^2 + \frac{1}{2} m \dot{x}_2^2$$

$$V = \frac{1}{2} k x_1^2 + \frac{1}{2} k (x_2 - x_1)^2 + \frac{1}{2} k x_2^2$$

$$L = T - V = \frac{1}{2} m (\dot{x}_1^2 + \dot{x}_2^2) - k (x_1^2 - x_1 x_2 + x_2^2)$$

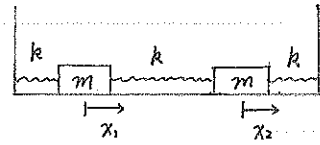
$$\frac{\partial L}{\partial \dot{x}_1} = m \dot{x}_1, \quad \frac{\partial L}{\partial x_1} = -2kx_1 + kx_2$$

$$\frac{d}{dt} \left(\frac{\partial L}{\partial \dot{x}_1} \right) - \frac{\partial L}{\partial x_1} = 0 \Rightarrow m \ddot{x}_1 + 2kx_1 - kx_2 = 0 \Rightarrow \ddot{x}_1 = -\frac{2k}{m} x_1 + \frac{k}{m} x_2 \quad (1)$$

$$\frac{\partial L}{\partial \dot{x}_2} = m \dot{x}_2, \quad \frac{\partial L}{\partial x_2} = kx_1 - 2kx_2$$

$$\frac{d}{dt} \left(\frac{\partial L}{\partial \dot{x}_2} \right) - \frac{\partial L}{\partial x_2} = 0 \Rightarrow m \ddot{x}_2 - kx_1 + 2kx_2 = 0 \Rightarrow \ddot{x}_2 = \frac{k}{m} x_1 - \frac{2k}{m} x_2 \quad (2)$$

(1) and (2) are the same as Eqs. (1.8.24) and (1.8.25).



2.1.3

$$T = \frac{1}{2} m (\dot{r}^2 + r^2 \dot{\theta}^2 + r^2 \sin^2 \theta \dot{\phi}^2)$$

$$V = V(r)$$

$$L = T - V = \frac{1}{2} m (\dot{r}^2 + r^2 \dot{\theta}^2 + r^2 \sin^2 \theta \dot{\phi}^2) - V(r)$$

$$\frac{\partial L}{\partial \dot{r}} = m \dot{r}, \quad \frac{\partial L}{\partial r} = m r \dot{\theta}^2 + m r \sin^2 \theta \dot{\phi}^2 - \frac{\partial V(r)}{\partial r}$$

$$\frac{d}{dt} \left(\frac{\partial L}{\partial \dot{r}} \right) - \frac{\partial L}{\partial r} = 0 \Rightarrow \underline{m \ddot{r} - m r \dot{\theta}^2 - m r \sin^2 \theta \dot{\phi}^2 + \frac{\partial V(r)}{\partial r} = 0}$$

$$\frac{\partial L}{\partial \dot{\theta}} = m r^2 \dot{\theta}, \quad \frac{\partial L}{\partial \theta} = m r^2 \sin \theta \cos \theta \dot{\phi}^2$$

$$\frac{d}{dt} \left(\frac{\partial L}{\partial \dot{\theta}} \right) - \frac{\partial L}{\partial \theta} = 0 \Rightarrow \underline{m r^2 \ddot{\theta} + 2 m r \dot{r} \dot{\theta} - m r^2 \sin \theta \cos \theta \dot{\phi}^2 = 0}$$

$$\frac{\partial L}{\partial \dot{\phi}} = m r^2 \sin^2 \theta \dot{\phi}, \quad \frac{\partial L}{\partial \phi} = 0$$

$$\frac{d}{dt} \left(\frac{\partial L}{\partial \dot{\phi}} \right) - \frac{\partial L}{\partial \phi} = 0 \Rightarrow \frac{d}{dt} (m r^2 \sin^2 \theta \dot{\phi}) = 0$$

$$\Rightarrow m r^2 \sin^2 \theta \dot{\phi} = \ell \quad (\ell \text{ is a constant})$$

$$\Rightarrow \underline{\underline{\dot{\phi} = \frac{\ell}{m r^2 \sin^2 \theta}}}$$

Note: $T = \frac{1}{2} m (\dot{x}^2 + \dot{y}^2 + \dot{z}^2)$

$$\begin{cases} x = r \sin \theta \cos \phi \\ y = r \sin \theta \sin \phi \\ z = r \cos \theta \end{cases} \Rightarrow \begin{cases} \dot{x} = \dot{r} \sin \theta \cos \phi - r \dot{\theta} \cos \theta \cos \phi - r \dot{\phi} \sin \theta \sin \phi \\ \dot{y} = \dot{r} \sin \theta \sin \phi + r \dot{\theta} \cos \theta \sin \phi + r \dot{\phi} \sin \theta \cos \phi \\ \dot{z} = \dot{r} \cos \theta - r \dot{\theta} \sin \theta \end{cases}$$

$$\begin{aligned}
T &= \frac{1}{2} m \left[\dot{r}^2 \sin^2 \theta \cos^2 \phi + \dot{r}^2 \dot{\theta}^2 \cos^2 \theta \cos^2 \phi + \dot{r}^2 \dot{\phi}^2 \sin^2 \theta \sin^2 \phi + 2 \dot{r} \dot{\theta} \sin \theta \cos \theta \cos^2 \phi \right. \\
&\quad - 2 \dot{r} \dot{\phi} \sin^2 \theta \cos \phi \sin \phi - 2 \dot{r}^2 \dot{\theta} \dot{\phi} \sin \theta \cos \theta \sin \phi \cos \phi + \dot{r}^2 \sin^2 \theta \sin^2 \phi + \\
&\quad \dot{r}^2 \dot{\theta}^2 \cos^2 \theta \sin^2 \phi + \dot{r}^2 \dot{\phi}^2 \sin^2 \theta \cos^2 \phi + 2 \dot{r} \dot{\theta} \sin \theta \cos \theta \sin^2 \phi + 2 \dot{r} \dot{\phi} \sin^2 \theta \sin \phi \cos \phi \\
&\quad \left. + 2 \dot{r}^2 \dot{\theta} \dot{\phi} \sin \theta \cos \theta \sin \phi \cos \phi + \dot{r}^2 \cos^2 \theta + \dot{r}^2 \dot{\theta}^2 \sin^2 \theta - 2 \dot{r} \dot{\theta} \sin \theta \cos \theta \right] \\
&= \frac{1}{2} m (\dot{r}^2 + \dot{r}^2 \dot{\theta}^2 + \dot{r}^2 \dot{\phi}^2 \sin^2 \theta)
\end{aligned}$$

2.5.1

$$\begin{aligned}
p_s &= \frac{\partial T}{\partial \dot{\theta}_s} = \sum_i \sum_j T_{ij}(\theta) \dot{\theta}_i \delta_{js} + \sum_i \sum_j T_{ij}(\theta) \delta_{is} \dot{\theta}_j \\
&= \sum_i T_{is}(\theta) \dot{\theta}_i + \sum_j T_{sj}(\theta) \dot{\theta}_j \\
\sum_s p_s \dot{\theta}_s &= \sum_s \sum_i T_{is}(\theta) \dot{\theta}_i \dot{\theta}_s + \sum_s \sum_j T_{sj}(\theta) \dot{\theta}_s \dot{\theta}_j \\
&= T + T \\
&= 2T.
\end{aligned}$$

2.5.2

$$L = \frac{1}{2} m \dot{x}^2 - \frac{1}{2} k x^2$$

$$p = \frac{\partial L}{\partial \dot{x}} = m \dot{x}$$

$$H = \dot{x} p - L = \frac{p^2}{2m} + \frac{1}{2} k x^2$$

Since L is not an explicit function of t , H is a conservative.

Set $H = E$, where E is a constant, we have

$$\begin{aligned}
\frac{1}{2} k x^2 + \frac{p^2}{2m} &= E \\
\Rightarrow \frac{x^2}{\frac{2E}{k}} + \frac{p^2}{2mE} &= 1.
\end{aligned}$$

Therefore, if we set $a^2 = 2E/k$, $b^2 = 2mE$, we have

$$\left(\frac{x}{a}\right)^2 + \left(\frac{p}{b}\right)^2 = 1.$$

2.5.3

$$L = \frac{1}{2} m (\dot{x}_1^2 + \dot{x}_2^2) - k (x_1^2 + x_2^2 - x_1 x_2)$$

$$p_1 = \frac{\partial L}{\partial \dot{x}_1} = m \dot{x}_1 \Rightarrow \dot{x}_1 = \frac{p_1}{m}$$

$$p_2 = \frac{\partial L}{\partial \dot{x}_2} = m \dot{x}_2 \Rightarrow \dot{x}_2 = \frac{p_2}{m}$$

$$H = \dot{x}_1 p_1 + \dot{x}_2 p_2 - L$$

$$= \frac{p_1^2}{2m} + \frac{p_2^2}{2m} + k(x_1^2 + x_2^2 - x_1 x_2)$$

$$\begin{cases} \dot{x}_1 = \frac{\partial H}{\partial p_1} = \frac{p_1}{m} & ① \\ \dot{p}_1 = -\frac{\partial H}{\partial x_1} = -2kx_1 + kx_2 & ② \end{cases}$$

From ①, we know $p_1 = m\dot{x}_1$. Therefore, $\dot{p}_1 = m\ddot{x}_1$. Substitute it into ②, we have $m\ddot{x}_1 = -2kx_1 + kx_2$.

$$\Rightarrow \ddot{x}_1 = -\frac{2k}{m}x_1 + \frac{k}{m}x_2. \quad ③$$

$$\begin{cases} \dot{x}_2 = \frac{\partial H}{\partial p_2} = \frac{p_2}{m} & ④ \\ \dot{p}_2 = -\frac{\partial H}{\partial x_2} = -2kx_2 + kx_1 & ⑤ \end{cases}$$

From ④, we know $p_2 = m\dot{x}_2$. Therefore, $\dot{p}_2 = m\ddot{x}_2$. Substitute it into ⑤, we have $m\ddot{x}_2 = -2kx_2 + kx_1$.

$$\Rightarrow \ddot{x}_2 = \frac{k}{m}x_1 - \frac{2k}{m}x_2. \quad ⑥$$

2.7.1

$$\{\omega, \lambda\} = \sum_i \left(\frac{\partial \omega}{\partial q_i} \frac{\partial \lambda}{\partial p_i} - \frac{\partial \omega}{\partial p_i} \frac{\partial \lambda}{\partial q_i} \right) = - \sum_i \left(\frac{\partial \lambda}{\partial q_i} \frac{\partial \omega}{\partial p_i} - \frac{\partial \lambda}{\partial p_i} \frac{\partial \omega}{\partial q_i} \right) = -\{\lambda, \omega\}.$$

$$\begin{aligned} \{\omega, \lambda + \sigma\} &= \sum_i \left(\frac{\partial \omega}{\partial q_i} \frac{\partial (\lambda + \sigma)}{\partial p_i} - \frac{\partial \omega}{\partial p_i} \frac{\partial (\lambda + \sigma)}{\partial q_i} \right) \\ &= \sum_i \left[\frac{\partial \omega}{\partial q_i} \left(\frac{\partial \lambda}{\partial p_i} + \frac{\partial \sigma}{\partial p_i} \right) - \frac{\partial \omega}{\partial p_i} \left(\frac{\partial \lambda}{\partial q_i} + \frac{\partial \sigma}{\partial q_i} \right) \right] \\ &= \sum_i \left[\left(\frac{\partial \omega}{\partial q_i} \frac{\partial \lambda}{\partial p_i} - \frac{\partial \omega}{\partial p_i} \frac{\partial \lambda}{\partial q_i} \right) + \left(\frac{\partial \omega}{\partial q_i} \frac{\partial \sigma}{\partial p_i} - \frac{\partial \omega}{\partial p_i} \frac{\partial \sigma}{\partial q_i} \right) \right] \\ &= \sum_i \left(\frac{\partial \omega}{\partial q_i} \frac{\partial \lambda}{\partial p_i} - \frac{\partial \omega}{\partial p_i} \frac{\partial \lambda}{\partial q_i} \right) + \sum_i \left(\frac{\partial \omega}{\partial q_i} \frac{\partial \sigma}{\partial p_i} - \frac{\partial \omega}{\partial p_i} \frac{\partial \sigma}{\partial q_i} \right) \\ &= \{\omega, \lambda\} + \{\omega, \sigma\}. \end{aligned}$$

$$\begin{aligned} \{\omega, \lambda \sigma\} &= \sum_i \left[\frac{\partial \omega}{\partial q_i} \frac{\partial (\lambda \sigma)}{\partial p_i} - \frac{\partial \omega}{\partial p_i} \frac{\partial (\lambda \sigma)}{\partial q_i} \right] \\ &= \sum_i \left[\lambda \frac{\partial \omega}{\partial q_i} \frac{\partial \sigma}{\partial p_i} + \frac{\partial \omega}{\partial q_i} \frac{\partial \lambda}{\partial p_i} \sigma - \lambda \frac{\partial \omega}{\partial p_i} \frac{\partial \sigma}{\partial q_i} - \frac{\partial \omega}{\partial p_i} \frac{\partial \lambda}{\partial q_i} \sigma \right] \\ &= \lambda \sum_i \left(\frac{\partial \omega}{\partial q_i} \frac{\partial \sigma}{\partial p_i} - \frac{\partial \omega}{\partial p_i} \frac{\partial \sigma}{\partial q_i} \right) + \sum_i \left(\frac{\partial \omega}{\partial q_i} \frac{\partial \lambda}{\partial p_i} - \frac{\partial \omega}{\partial p_i} \frac{\partial \lambda}{\partial q_i} \right) \sigma \\ &= \lambda \{\omega, \sigma\} + \{\omega, \lambda\} \sigma. \end{aligned}$$